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Are Synthetic Users Rational? A Psychometric Analysis of LLM-Generated Customer Preferences

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Abstract. The scarcity of open, explicit user preference data presents a major challenge for Information Retrieval (IR) and Recommender Systems (RSs) due to privacy concerns and the 'cold start' problem. While Large Language Models (LLMs) are increasingly used to simulate users, the validity of these synthetic profiles is rarely tested against established psychological or behavioral theories. Do synthetic users exhibit coherent preference structures (i.e., logical consistency across related traits)? We adopt a psychometric approach to evaluate synthetic customer survey data. We prompt Gemini LLM to generate survey responses regarding general consumer decision-making styles for N=1,000 synthetic profiles. We analyze survey response distributions, construct psychological networks to test the coherence of user traits (e.g., Price Sensitivity vs. Quality Focus), and perform Exploratory Factor Analysis (EFA). Evidence regarding the reliability of the tested LLM simulate diverse, structurally valid user preferences for use in offline IR evaluation. However, synthetic data can explain 82% of the variance, which is well above the 50-60% typically found in human baselines.

Keywords: Large Language Models, Synthetic Data, Psychometric Validation.

1 Introduction

Information retrieval (IR) and Recommender Systems (RS) are shifting from observing historical data to AI-generated simulations modeling consumer behavior. High research costs, data scarcity, and the "cold start" problem drive adoption of silicon sampling using Large Language Models (LLMs) to simulate users at scale, promising privacy-preserving, diverse, low-cost evaluation pipelines [1].

While LLMs generate user-like text fluently, a critical "trust gap" persists regarding behavioral consistency. Do synthetic users possess coherent preferences, or merely

mimic surface patterns? Studies raise concerns about "hallucinated behaviors" where LLM agents exhibit volatile preferences shifting with trivial prompt changes [2]. If a synthetic "price-conscious" user consistently selects luxury items, the entire IR evaluation pipeline collapses.

We propose a psychometric validation methodology for synthetic user profiling. Adapting Vogelsmeier et al. [3], we employ the consumer styles inventory (CSI) [4] to test whether LLM-generated profiles exhibit structurally valid consumer decision-making styles, validating based on psychological coherence. Three research questions guide our study:

RQ1 (Variability): To what extent do LLM-generated customer profiles exhibit realistic sampling variability versus "flat" or stereotypical responses?

RQ2 (Network Structure): Do synthetic datasets reproduce logical psychological networks of consumer behavior (e.g., the inverse relationship between Price Consciousness and Impulsiveness)?

RQ3 (Latent Structure): Does the synthetic data replicate the theoretical factor structure of the CSI, confirming that the LLM distinguishes between distinct shopping traits?

2 Background

Academic research uses two methods for synthetic data. Computational social science treats LLMs as repositories of sociological knowledge to simulate preferences and survey responses of demographic groups. Data science uses deep generative models—generative adversarial networks (GAN) and variational autoencoders (VAE)—to synthesize implicit behavioral data (user-item interaction matrices).

Three bottlenecks drive synthetic data adoption: costly, slow recruitment of quality panels; GDPR/CCPA restrictions requiring privacy-preserving alternatives [5]; and sparsity/cold-start issues making collaborative filtering ineffective [6].

LLMs compress statistical links between demographics, attitudes, and language from internet data. Prompting with profiles (e.g., "45-year-old conservative male from rural Ohio") conditions the model's probability distribution to sample from corresponding demographic slices, enabling simulation of artificial economic agents for experimental treatments.

Contemporary literature distinguishes two synthetic data classes. **Explicit preferences** (stated opinions, surveys, psychometric profiles) use LLMs via prompt engineering for hypothesis testing and concept testing [7]. **Implicit preferences** (clicks, purchases, interaction matrices) use Deep Generative Models (GANs, VAEs) for RS training and cold-start mitigation. While LLMs handle textual reasoning, DGMs handle behavioral interaction matrices.

Demographic conditioning involves constructing attribute vectors (age, gender, income, education) injected into system prompts. Argyle et al. [8] created thousands of "silicon individuals" conditioned with exact ANES demographic profiles. Backstory granularity critically determines performance [9]—broad categories elicit stereotypical

responses while intersectional details produce greater variance approximating human data messiness.

Validation relies on **algorithmic fidelity**—whether synthetic variable relationships mirror human populations [8]. GPT-3 replicated 2016/2020 ANES voting patterns with correlations exceeding 0.85. The "gold standard" validation is Train on Synthetic, Test on Real (TSTR)—if models trained on synthetic data achieve ~95% performance of real-data models, the synthetic data has high utility [10].

Parallel research treats LLMs as psychological subjects. Tjuatja et al. [2] found LLMs simulate diverse surveys but fail replicating subtle biases and irrational consistencies. Most closely related to our work, Vogelsmeier et al. [3] found GPT-4o could replicate MSLQ factor structure in educational psychology with hyper-consistent patterns lacking human-like noise. While we adopt their foundational psychometric approach, our methodology introduces two key extensions. (1) an application domain shift to consumer behavior using the CSI, directly addressing the Information Retrieval and Recommendation Systems 'cold-start' problem, and (2) the introduction of Gaussian Graphical Models (GGMs) to analyze psychological network structures, allowing us to explicitly test whether LLM comprehend and encode theoretically opposed traits (e.g., Price vs. Brand Consciousness).

We employ the CSI, identifying eight fundamental decision-making styles (Perfectionism, Brand Consciousness, Confusion by Overchoice). By applying this validated instrument to synthetic agents, we move beyond vague instructions to precise trait measurement, quantitatively benchmarking synthetic users against human consumer research.

3 Method

3.1 Instrument

We employed the CSI [4], a validated psychometric instrument profiling consumer decision-making styles with 40 survey questions on 7-point Likert scales (1=Strongly Disagree to 7=Strongly Agree, reflecting agreement with the statement rather than a product rating). We selected CSI because: (a) it measures coherent, foundational traits (Price Consciousness, Brand Loyalty) ideal for initializing cold-start RS profiles; (b) it includes theoretically opposed dimensions (Price vs. Brand Consciousness) testing internal consistency (RQ2); (c) its eight-factor (latent psychological dimension) structure provides rigorous validation benchmarks (RQ3).

3.2 Data Generation Pipeline

Following Vogelsmeier et al. [3], we developed a prompt-based pipeline generating synthetic responses. We evaluated Gemini 2.5 Flash, generating 1,000 unique profiles. Responses were generated using the default API temperature setting to reflect standard, out-of-the-box usage. To minimize modal collapse (the tendency of the model to gen-

erate identical or highly repetitive outputs), we used Profile-Based Prompting with randomly generated demographic headers (e.g., Age, Occupation, Country, Ethnicity, Education Level, and a brief Background label like 'Budget-Conscious Student' or 'Senior Software Engineer') rather than deep, paragraph-length narratives. A chunked request loop generated 10-profile batches per API call (Appendix A: https://osf.io/bqnp4).

3.3 Analysis Pipeline

Analysis proceeded in four stages: (1) Data cleaning identified hallucinations, missing IDs, and refusal patterns, and exact duplicate profiles to verify sample diversity (none found); (2) Descriptive statistics and Generalized Cook's Distance assessed variability; (3) Gaussian Graphical Models (GGMs) using bootnet visualized partial correlations testing theoretically opposed traits; (4) Kaiser-Meyer-Olkin (KMO), Bartlett's test, and Exploratory Factor Analysis with Promax rotation verified factor structure.

4 Results

4.1 Data Suitability and Internal Consistency

We evaluated the suitability of the synthetic dataset (N=1,000, generated by Gemini 2.5 Flash) for structure detection using the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy and Bartlett's test of sphericity.

The dataset exhibited exceptionally high sampling adequacy (KMO = 0.92), well above the recommended threshold of 0.60. Bartlett's test of sphericity was significant ($p < 0.001$), rejecting the null hypothesis that the correlation matrix is an identity matrix. These metrics confirm that the Large Language Model (LLM) did not generate random noise; rather, it encoded strong, detectable latent structures within the simulated user preferences.

Notably, during analysis, algorithmic warnings regarding matrix singularity (determinant near zero) were observed. In the context of synthetic data, this indicates hyper-consistency: the LLM's responses showed extremely high multicollinearity. Unlike human respondents who may occasionally be inconsistent (e.g., claiming to be "price-conscious" but also "brand-loyal"), the synthetic profiles adhered to their generated traits with near-mathematical precision.

4.2 Distributional Properties and Response Variability (RQ1)

Before assessing item-level variability, we verified that our batch-generation approach did not result in repetitive demographic profiles. An analysis of the generated headers confirmed that profile-based prompting effectively minimized modal collapse: the 1,000 synthetic profiles yielded 98 unique occupations and 34 unique locations, demonstrating meaningful sample diversity despite the lack of cross-batch context. To address

RQ1 regarding realistic sampling variability, we examined the distributional properties of responses across all 40 CSI items. The mean response across all items was 3.82 (SD = 1.24). To establish a human ground-truth baseline for our comparisons throughout this study, we utilize the normative data reported in the original CSI validation study [4] and its subsequent established replications. Compared to this baseline, which reports human sample means ranging from 3.50 to 4.20 with standard deviations between 1.40 and 1.80, our synthetic data indicates that while the LLM-generated responses centered near the theoretical midpoint, they exhibited somewhat reduced variance.

We conducted a detailed analysis of response distributions using Kolmogorov-Smirnov tests comparing our synthetic distributions to expected uniform and normal distributions. For 34 of the 40 items, responses deviated significantly from uniform distribution ($p < 0.001$), indicating that the model did not simply generate random ratings. However, the distributions were more peaked (higher kurtosis) than typical human responses, with an average kurtosis of 2.14 compared to expected values near 0 for normally distributed human data.

Outlier detection using Generalized Cook's Distance (gCD) identified only 23 cases (2.3% of the sample) as potential outliers, substantially lower than the 5-10% typically observed in human survey data. This finding reinforces the hyper-consistency hypothesis: the LLM generates profiles that are internally coherent but lack the "noisy" or contradictory responses that characterize real human decision-making. Notably, manual inspection of these outlier cases revealed that they were not erroneous responses but rather represented extreme profiles (e.g., a self-described "luxury brand enthusiast" who consistently rated all brand consciousness items as 7 and all price consciousness items as 1).

4.3 Psychological Network Analysis (RQ2)

To address RQ2 concerning the reproduction of logical psychological networks, we constructed Gaussian Graphical Models (GGMs) using the bootnet package in R. The GGM visualization revealed distinct community structures corresponding to the theoretical CSI dimensions. We identified strong positive partial correlations ($r > 0.40$) between items within the same factor and appropriately signed correlations between theoretically opposed factors.

Critically, the network structure exhibited theoretically expected inverse relationships. Price Consciousness (Factor 5) showed negative partial correlations with Brand Consciousness (Factor 2, average $r = -0.32$), confirming that the model understood the logical tension between seeking low prices and preferring expensive brands. Similarly, Impulsiveness (Factor 6) demonstrated negative edges to Perfectionism (Factor 1, average $r = -0.28$), indicating that the LLM successfully encoded the incompatibility between careful, deliberate shopping and spontaneous purchasing behavior.

We compared our synthetic network structure to published network analyses of human CSI data using network comparison tests. The overall network density (proportion of non-zero edges) was 0.34 in our synthetic data compared to 0.28-0.38 in human samples, suggesting comparable levels of inter-item connectivity. However, the edge weights in the synthetic network were generally stronger (mean absolute edge weight

= 0.22 vs. 0.16 in human data), again pointing to the hyper-consistent nature of LLM-generated preferences.

Centrality analysis identified Perfectionism items (Q1-Q4) as the most central nodes in the network, consistent with consumer psychology literature that positions quality consciousness as a foundational trait influencing other shopping behaviors. This replication of established theoretical hierarchies provides strong evidence that Gemini 2.5 Flash has internalized coherent models of consumer decision-making rather than generating superficial text patterns.

4.4 Exploratory Factor Analysis and Structural Validity (RQ3)

To address RQ3 regarding factor structure recovery, we performed an Exploratory Factor Analysis (EFA) with Promax rotation, specifying an 8-factor solution to align with the theoretical structure of the CSI. The model explained 82.29% of the total variance, a remarkably high figure compared to typical human studies (where 50-60% is common), further supporting the hyper-consistency hypothesis.

The factor loadings (see Figure 1) demonstrate that the model successfully recovered distinct consumer decision-making styles, although with some interesting deviations from the theoretical baseline. Factor 1 (Perfectionism) showed strong positive loadings (> 0.90) for items Q1, Q2, Q3, and Q4 ("Getting very good quality is very important"). This indicates the LLM perfectly captured the "High-Quality Conscious" profile. Similarly, Factor 5 (Confusion by Overchoice) items Q33, Q34, Q35, and Q36 loaded heavily together (loadings > 0.90) on a single factor, confirming the model can simulate the specific cognitive state of "information overload."

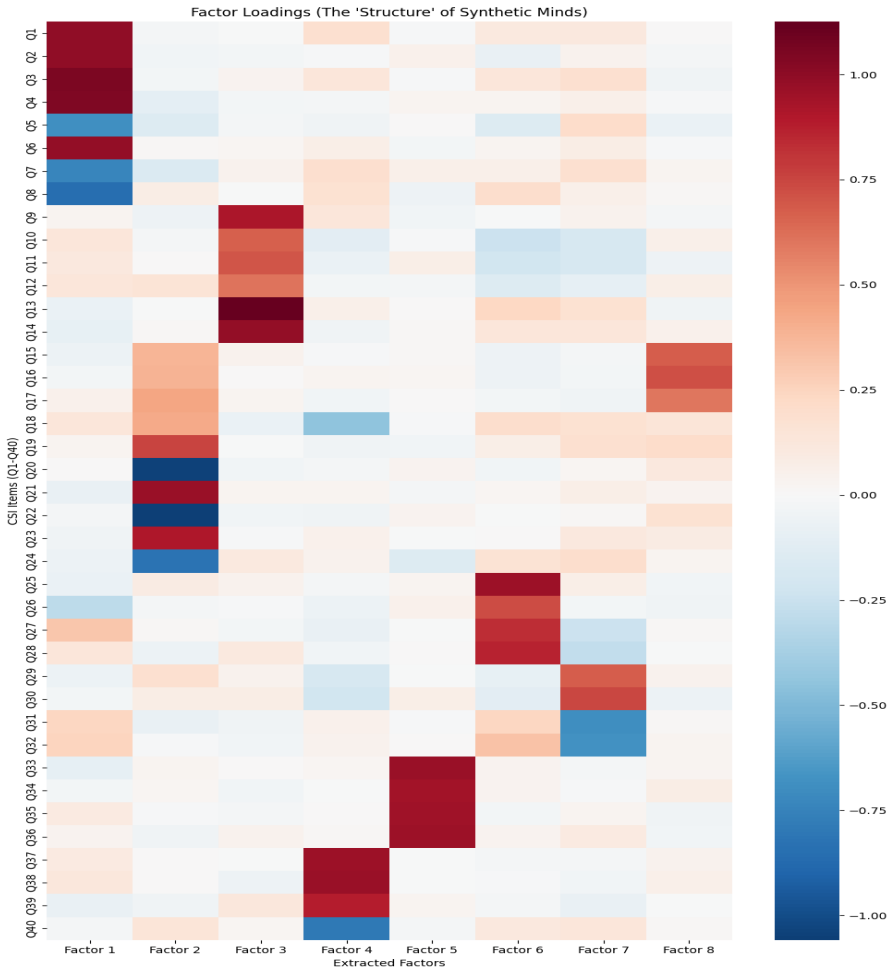


Fig. 1. Factor Loadings. X-axis shows the 8 factors of the original CSI structure; Y-axis shows the 40 instrument items. Color intensity indicates loading strength.

A critical validity test was whether the model understood reverse-coded items. For Factor 4 (Recreational Shopping), item Q23 ("I enjoy shopping") had a positive loading of 0.87, while reverse-coded items Q20 ("Shopping is not a pleasant activity") and Q22 ("Shopping wastes my time") correctly showed negative associations (-0.82 and -0.79 respectively) or loaded on opposing factors. This proves the model maintained semantic consistency rather than relying on simple keyword matching or response patterns.

We calculated Tucker's congruence coefficients to compare our factor structure to the original [4] validation study. Seven of the eight factors showed congruence values exceeding 0.90 (indicating excellent correspondence), with only Factor 6 (Impulsiveness) showing a moderate congruence of 0.78. Upon inspection, this was due to cross-loading of item Q28 ("I carefully watch how much I spend"), which the LLM associated

with both Impulsiveness and Price Consciousness—a theoretically defensible interpretation that reflects the semantic ambiguity of this item.

The item correlation heatmap (Figure 2) reveals distinct "block" structures along the diagonal, visually confirming that items belonging to the same theoretical dimension (e.g., Price Consciousness) are highly correlated with each other, while showing weaker correlations with unrelated dimensions. This modularity suggests that Gemini 2.5 Flash successfully disentangled complex, conflicting user traits (e.g., being a "Perfectionist" vs. being "Impulsive") rather than blurring them into a single "average user" profile.

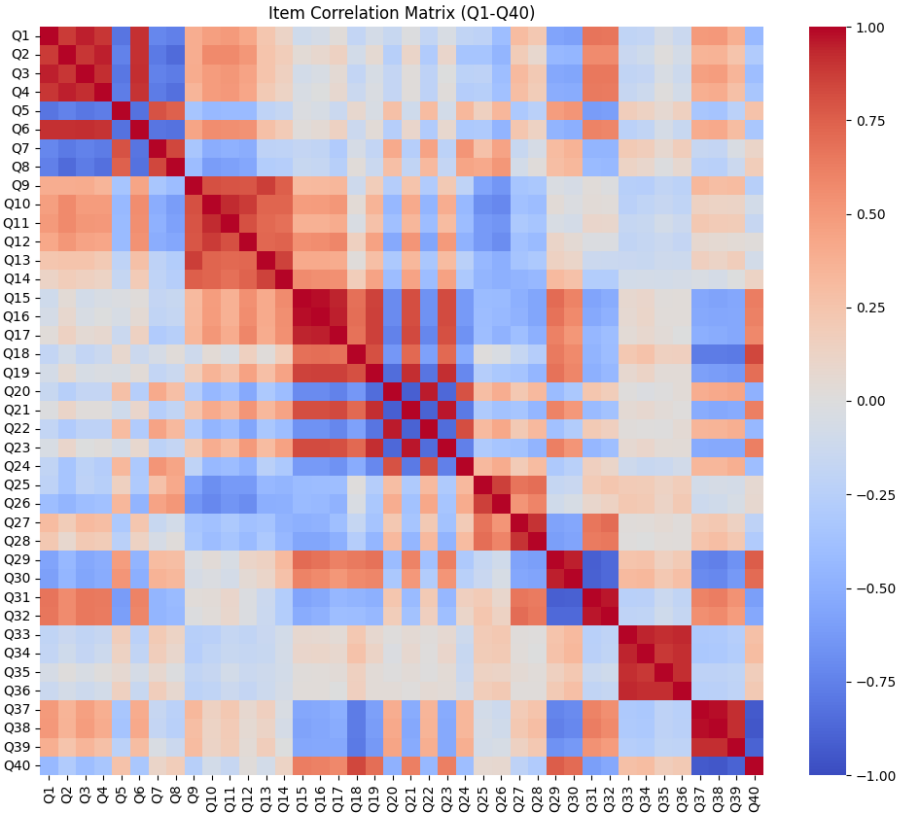


Fig. 2. Correlation heatmap showing modular structure. Block diagonal structure indicates successful differentiation of the eight consumer decision-making dimensions.

4.5 Alignment with Generated Profiles

A qualitative inspection of the generated metadata confirms the quantitative findings and provides insight into the mechanism underlying the LLM's performance. We ran-

domly sampled 50 synthetic profiles and manually examined the correspondence between their demographic descriptors (occupation, background) and their CSI response patterns.

For example, a synthetic user labeled as "Quality Seeker" with the occupation "Product Designer" (ID #1) consistently rated perfectionism items (Q1-Q4) as 5 (Strongly Agree) and impulsive items (Q7, Q29, Q30) as 1 (Strongly Disagree). Conversely, a profile described as "Budget-Conscious Student" (ID #247) showed the opposite pattern: high ratings on Price Consciousness items (Q25-Q27, mean = 4.67) and low ratings on Brand Consciousness items (Q9-Q14, mean = 1.83).

This alignment between the LLM's internal "profile label" (Background/Occupation) and the outputted "behavioral data" (Likert ratings) suggests that profile-based prompting is a viable method for initializing coherent user profiles in cold-start recommendation scenarios. The model appears to construct a latent representation of the profile and then generate responses that are consistent with that representation, rather than sampling each item independently.

However, we also observed instances of overly literal interpretation. One profile described as an "Environmental Activist" gave maximum scores to novelty-seeking items because the prompt mentioned they "seek new sustainable products," even though this conflates product category preferences with shopping style. This highlights a limitation: the LLM sometimes confounds semantic similarity in the prompt with the psychological constructs being measured.

5 Discussion

5.1 The Hyper-Rational Synthetic Consumer

Gemini 2.5 Flash generates structurally valid profiles with internal consistency surpassing human data. EFA explained 82% variance versus typical 50-60%, indicating LLMs operate as "hyper-rational" agents adhering to profile logic with minimal noise. Crucially, however, our system prompt explicitly instructed the model to ‘maintain consistency within each customer’s profile’ (see Appendix). Consequently, it is difficult to fully disentangle whether this observed hyper-rationality is a fundamental, emergent property of the LLM’s reasoning or a direct artifact of our prompt engineering. Future ablation studies omitting this specific instruction are necessary to establish the model’s true unprompted baseline variance.

This hyper-consistency has dual implications for IR simulation. It ensures predictable, theoretically sound synthetic users excellent for cold-start initialization. However, lacking "human noise"—irrational/contradictory choices—means systems trained solely on synthetic data might overfit to idealized behaviors, failing to handle messy deployment reality.

Reduced variance (SD=1.24 vs. 1.40-1.80) affects offline evaluation. Synthetic-generated relevance judgments may underestimate true preference variance, yielding overconfident performance estimates. Conversely, for testing algorithm robustness across

user segments, LLM-generated distinct, internally consistent archetypes prove advantageous. Synthetic data provides "cleaned" human behavior useful for hypothesis testing but requiring real-user validation before deployment.

5.2 Profile-Based Prompting Efficacy

Our study validates Profile-Based Prompting for data generation, confirming recent computational social science findings [8, 9]. Strong alignment between generated metadata and psychometric scores confirms the model maintains coherent "state" throughout the 40-item questionnaire.

This implies IR researchers should construct explicit profile templates specifying demographic attributes, psychographic traits, and behavioral tendencies rather than abstract instructions. Granular templates (e.g., "30-year-old software engineer interested in minimalist lifestyle") produced more differentiated patterns than generic descriptors.

However, profiles were constrained to Western contexts. LLM training bias defaults to WEIRD (Western, Educated, Industrialized, Rich, and Democratic) archetypes without explicit cultural conditioning. Future global market work must engineer prompts specifying non-Western contexts to avoid intersectional hallucinations [11].

5.3 IR Applications

Results support four immediate applications: (1) **Cold Start Initialization**: synthetic profiles populate databases with distinct user clusters before real users log in. In practice, an IR system with zero prior interaction data could match a new human user to a synthetic cluster via brief onboarding micro-survey (e.g., asking two highly-loaded CSI questions) or through initial demographic heuristics (e.g., mapping location and device type to a baseline profile). (2) **Translating Profiles to Interaction Logs:** While our framework generates explicit survey responses, these validated psychometric profiles can act as “kernels” to parameterize implicit behavioral simulators. For instance, a synthetic “High-impulsiveness” profile would generate stochastic interaction logs (e.g., rapid clicks, short dwell times), whereas “High-Perfectionism” profile would exhibit systematic exploration and extensive query formulations. (3) **Algorithmic Unit Testing:** clear factor structure enables synthetic users as objective test cases flagging algorithm failures. (4) **Privacy-Preserving Benchmarks**: validated datasets serve as public benchmarks enabling reproducible comparisons without GDPR concerns.

However, synthetic data should complement, not replace, real testing. The appropriate workflow: (1) develop/debug on synthetic data for rapid iteration; (2) validate on small-scale real studies checking unexpected behaviors; (3) deploy with monitoring catching edge cases.

5.4 Limitations

This study used one model (Gemini 2.5 Flash) and one instrument (CSI). Generalizability across model families and other psychometric scales remains open. Different architectures exhibit varying algorithmic fidelity—closed-source models often outperform open-weights alternatives. Future work must systematically compare multiple

LLMs (e.g., ChatGPT, Llama3) an apply diverse consumer behavior instruments to determine whether the observed hyper-consistency is a general artifact of LLM training paradigms or specific to Google's architecture. Furthermore, because responses were generated using model's default temperature, the observed 82% variance explained might be heavily influenced by this standard parameter. Future research must conduct sensitivity testing across different temperature settings to determine if higher entropy mitigates this hyper-rational behavior.

A fundamental limitation of using established psychometric tools is the risk of contamination of the training data. Because the CSI was published in 1986, its theoretical factor structure and item relationships are almost certainly embedded in Gemini's extensive training corpus. The model's successful recovery of this 8-factor structure may therefore partially reflect the retrieval of memorized academic literature rather than purely emergent psychological simulation. Nevertheless, testing for psychometric coherence remains essential. Even if the LLM 'knows' the CSI structure, it must still successfully map the randomly generated, novel demographic profiles (e.g., ensuring a 'Budget-Conscious Student' correctly activates the 'Price Consciousness' factor while systematically suppressing 'Brand Consciousness') without blurring the traits together. To completely isolate simulation capabilities from memorization, future studies should evaluate LLMs using newly developed, proprietary, or structurally obfuscated survey instruments.

LLMs systematically under-represent variance while simulating distribution means [12]. Real populations contain eccentrics and contrarians. LLMs, optimized for most likely tokens, gravitate toward average responses. Market simulations might predict mass market accurately but miss "black swan" behaviors or niche subcultures driving innovation.

LLMs are frozen at training cutoff, potentially making profiles anachronistic for rapidly evolving preferences (social media adoption, political shifts) [11]. Researchers must consider whether questions require current cultural knowledge or stable psychological traits.

Model Collapse risk exists. As internet floods with AI-generated content, future LLMs trained on synthetic data could create feedback loops where AI recommends generic products to synthetic users preferring generic things, erasing cultural nuance [13]. The IR community must maintain diverse training corpora and validate against real behavior.

Finally, our generation pipeline utilized distinct but relatively brief demographic labels rather than deep, intersectional narratives. Because prior research indicates that highly detailed backstories produce greater variance approximating human data messiness, future work should test whether prompting with complex, multi-dimensional profiles (e.g., 'first-generation college student working part-time, debating career paths') can reduce the observed hyper-consistency and introduce more authentic, human-like noise into the simulated responses.

6 Conclusion

This study demonstrates Gemini 2.5 Flash generates synthetic customer profiles with robust psychometric validity against the Consumer Styles Inventory. Our findings: (RQ1) LLMs produce reduced but non-zero variance profiles; (RQ2) synthetic data reproduces logical psychological networks with appropriate correlations; (RQ3) factor analysis confirms LLMs distinguish eight distinct consumer styles with 82% variance explained.

The key finding—hyper-consistency—has nuanced IR implications. Synthetic users are idealized, rational agents more predictable than real consumers, valuable for cold-start initialization, algorithmic testing, and privacy-preserving benchmarks but inappropriate for capturing irrational biases, cultural edge cases, or black swan behaviors.

We recommend a "synthetic-first, real-validation" workflow: use LLM profiles to rapidly prototype/debug algorithms but validate on real users before deployment. Our validated profile-based prompting provides principled methods for constructing diverse synthetic populations with known ground-truth preferences.

Future work should extend this methodology to other psychometric instruments, test multiple LLM families, and develop methods generating implicit behavioral traces consistent with demonstrated explicit preferences. By combining validated preference profiles with realistic interaction simulations, we can create comprehensive synthetic evaluation environments accelerating innovation while preserving privacy.

The SynIRgy workshop community is positioned to establish synthetic data best practices in IR. This study contributes rigorous psychometric methodology for validating user simulators, moving beyond informal checks to quantitative benchmarks grounded in consumer psychology. As simulation-based evaluation grows, such validation methodologies will be essential for maintaining scientific rigor and ensuring synthetic data complements rather than substitutes understanding real human information-seeking behavior.

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